ECEN-662: Homework 7

Due on April 12, 2026 at 11:59 PM

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Problem 1

Given a uniform random variable $U \sim \mathcal{U}([0,1])$, derive the inverse transform sampling function required to generate samples from a Rayleigh distribution with parameter σ:

$$f(x) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right), \quad x \geq 0.$$

Solution

Problem 2

Suppose we want to sample from a standard Normal distribution, but only the positive half (a truncated normal). The target density

$$f(x) \propto \exp(-x^2/2), \quad x > 0.$$

We will use an exponential distribution

$$g(x) = \lambda \exp(-\lambda x), \quad x \geq 0$$

as our proposal distribution.

(i) Find the optimal scaling constant M as a function of λ such that $f(x) \leq Mg(x)$ for all $x \geq 0$.

(ii) Analytically determine the optimal rate parameter λ that minimizes the rejection rate.

Solution

Problem 3

We want to estimate the probability of a standard normal random variable exceeding a threshold of 4:

$$\mathbb{P}(X > 4) = \int_4^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx.$$

(i) If we use standard Monte Carlo by drawing $N = 10^5$ samples from $X \sim \mathcal{N}(0, 1)$, what is the expected number of samples that will fall in the region $X > 4$?

(ii) Propose a shifted Gaussian importance sampling distribution

$$q(x) = \mathcal{N}(\mu, \sigma^2)$$

that would significantly reduce the variance of the estimate. Justify your choice of μ and write out the final Importance Sampling estimator equation.

Solution

Problem 4

Let the target distribution be an unnormalized symmetric bimodal distribution:

$$\pi(x) \propto \exp\left(-\frac{(x-3)^2}{2}\right) + \exp\left(-\frac{(x+3)^2}{2}\right).$$

We will use a Metropolis-Hastings algorithm with a Gaussian random walk proposal:

$$p(x, x') = \mathcal{N}(x'; x, \sigma_p^2).$$

Assume the current state of the chain is $x_t = 3.0$ (sitting precisely on the right peak).

(i) Calculate the exact acceptance ratio $\alpha(x_t, x')$ if the proposal draws a new state $x' = 0.0$ (the deep valley between the peaks).

(ii) Conceptually, what happens to the Markov Chain if the proposal variance σ_p^2 is set extremely small (e.g., 0.01)? What happens if it is set extremely large (e.g., 100)?

Solution